Hailemariam (Haile) Teshome (He/Him/His)

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EXPERIENCE

AI/ML Engineer Intern Jun 2026 - Dec 2026

Siemens Healthineers / Varian Medical Systems, Palo Alto, CA

- Building an agentic system that automates document-review workflows across medical-device development, orchestrating multi-step retrieval, cross-checking, and summarization over design-control documentation to generate requirements-traceability and compliance evidence for external audits.

Graduate AI/ML Researcher Nov 2025 - Present

Multimodal Medical AI & Oncology Labs, Bakar Computational Health Sciences Institute, UCSF, San Francisco, CA

Research Advisors: Dr. Julian Hong, Dr. Madhumita Sushil

- Develop longitudinal multimodal models integrating imaging, pathology, and EHR signals with SSM/Transformer architectures, enabling brain and prostate cancer diagnostics from full patient trajectories.
- Curate and preprocess large multimodal clinical datasets (imaging, free-text pathology, structured EHR) into reproducible pipelines for model training and benchmarking.
- Built a GRPO reinforcement-learning harness to post-train Qwen2.5-7B for radiation-oncology treatment planning with learned process reward models, rule-based verifiers, and NCCN-grounded evaluation.
- Isolated reward design as the primary driver of unsafe over-treatment in clinical RL: outcome- and process-level rewards improved NCCN guideline concordance, while a learned process reward model reduced it by 0.177 (95% CI [-0.207, -0.142]) vs SFT warm-start.

Graduate AI/ML Researcher/Engineer Oct 2025 - Present

UCSF / Brown University, San Francisco, CA

Advisors: Sarah Ackley, PhD; Christopher Seaman, MS, Thomas J. Hoffmann, PhD

- Architected a full-stack, open-source, AI-powered systematic-review platform (Python, FastAPI, React, TypeScript, LangChain) that searches, deduplicates, and screens thousands of papers across a multi-database search strategy (i.e. PubMed, arXiv, bioRxiv, Semantic Scholar, CORE) from a single interface.
- Engineered a parallel multi-agent LLM screening pipeline served using SGLang running 16 concurrent workers to score papers against PICO and inclusion/exclusion criteria; defined the evaluation methodology, achieving 0.849 AUROC and ~95% mean sensitivity on the 26-review SYNERGY benchmark.
- Built a human-in-the-loop review layer with AI-decision override, full audit trail, auto-generated PRISMA flow diagrams, and structured data exporting; coordinated collaborators across UCSF and Brown as first author on the resulting manuscripts (LM4Sci Workshop @ COLM 2026, accepted; JAMIA, under review).

AI/ML Engineer Intern May 2025 - Aug 2025

Genentech, gRED Clinical Development Informatics (Digital Transformation Office), South San Francisco, CA

- Deployed a multi-agent LLM system (LangChain, MCP, CrewAI) automating clinical-trial protocol quality checks, study-design review, and data validation, cutting manual review cycles and protocol amendments.
- Engineered a production SaaS platform (Python, FastAPI, Docker, PostgreSQL, Vue.js/React) for Clinical Science, Operations, and Informatics teams, wiring LLM agents to internal data sources via MCP for dynamic context retrieval; adopted by a 20-user team processing 30 protocols/month.
- Built an end-to-end ML observability stack with MLflow for experiment tracking across model versions, prompts, and datasets; Prometheus and Grafana for real-time latency, throughput, and error-rate monitoring.

Software Engineer Jul 2022 - May 2025

Amgen, Decision Sciences & Generative AI Teams, Thousand Oaks, CA (Remote)

- Co-architected a petabyte-scale drug-portfolio recommendation and HCP-segmentation system (Databricks, Spark, Python, AWS, SQL) with automated GitLab CI/CD, delivering 10K–100K+ recommendations weekly into commercial decision-support workflows; optimizations cut deployment failures 60%.
- Onboarded and mentored 20+ peer engineers onto the platform through knowledge-transfer sessions, recorded trainings, and documentation; established the team's CI/CD and code-review standards.
- Built an automated cost-auditing system monitoring weekly advertising-journey sends and spend at per-notebook granularity, with weekly email reports of week-over-week changes and monthly trends.
- Built RAG-based internal tooling, including a direct Jira-to-release-notes generator and an automated root-cause-analysis system for deployment failures, reducing mean time to resolution from 2 weeks to 1.5 days.

- Led GenAI adoption by deploying custom GPT-based tools, ran technical workshops, and authored internal guidelines for responsible model use.

Research Engineer Jun 2022 - Jul 2022

Harvard T.H. Chan School of Public Health, Global Health & Population, Boston, MA

- Optimized R-based COVID-19 population-estimate simulations and customized R packages to project needs, reducing runtime by 24%.
- Built data-aggregation and processing systems for large, multi-source COVID-19 datasets.
- Developed reports and dynamic D3.js visualizations of model findings for stakeholders and publications.

Machine Learning / Data Science Intern Aug 2018 - Jul 2019

UCLA Health / David Geffen School of Medicine, Bioinformatics & Health Analytics, Los Angeles, CA

- Developed end-to-end ETL pipelines in SQL and Python to process and clean large perioperative clinical datasets.
- Built in-database ML models (SQL, Python, R) for perioperative and anesthesiology outcomes analytics.
- Designed Tableau dashboards that translated complex datasets into actionable visualizations, helping clinicians and administrators identify trends and make data-driven decisions.

Undergraduate Researcher Sep 2016 - Sep 2019

Bio and Nano Photonics Laboratory, UCLA, Los Angeles, CA

Research Advisors: Dr. Aydogan Ozcan, Dr. Hyou-Arm Joung

- Engineered multiplexed paper-based biosensors for Lyme-disease detection and cell-motility analysis using photonics and image-processing techniques.
- Built a MATLAB image-analysis algorithm achieving 97% accuracy in quantifying sensor-antigen interactions.
- Developed a deep-learning diagnostic algorithm for Lyme disease (96.5% specificity, 85.7% sensitivity); work contributed to two peer-reviewed journal publications.

EDUCATION

M.S. Health Data Science 2026

University of California, San Francisco, San Francisco, CA

Teaching Assistant: Bayesian Methods and Gaussian Processes (DATASCI 226)

UCSF-Stanford Exchange Scholar 2025 - 2026

Stanford University, Stanford, CA

Coursework: Foundation Models in Healthcare, Deep Reinforcement Learning

B.S. Atmospheric and Oceanic Science 2021

University of California, Los Angeles, Los Angeles, CA

TECHNICAL SKILLS

Languages: Python, SQL, C++, R, JavaScript, Bash, MATLAB

LLM / GenAI: LangChain, LangGraph, RAG, LlamaIndex, pgvector, FAISS, Chroma, MCP, CrewAI, SGLang/Ollama

ML / Deep Learning: PyTorch, Hugging Face, scikit-learn, numpy, pandas

MLOps / Infrastructure: AWS, Docker, Databricks, Spark, Airflow, MLflow, GitLab CI/CD, Prometheus, Grafana, Arize, LangSmith, Modal

Full-Stack: React, Vite, FastAPI, Flask, Streamlit, PostgreSQL, Tableau, D3.js

SELECTED PUBLICATIONS

Teshome H, Hoffmann T, Seaman C*, Ackley S*. *Evidence Engine: An AI-Enabled, Human-in-the-Loop Informatics Tool for Rapid Systematic Reviews.* LM4SCI Workshop @ COLM **[Accepted]**, Journal of the American Medical Informatics Association **[Under Review]**.

Teshome H. *Who Gets the Second Opinion? Selective Prediction Redistributes Demographic Risk in Medical AI.* Machine Learning for Health (ML4H) Proceedings Track **[Accepted]**.

Joung HA, Ballard ZS, Wu J, Tseng DK, **Teshome H**, Zhang L, Horn EJ, Arnaboldi PM, Dattwyler RJ, Garner OB, Di Carlo D, Ozcan A. *Point-of-Care Serodiagnostic Test for Early-Stage Lyme Disease Using a Multiplexed Paper-Based Immunoassay and Machine Learning.* **ACS Nano, 2020.**

Joung HA, Ballard ZS, Ma A, Tseng DK, **Teshome H**, Burakowski S, Garner OB, Di Carlo D, Ozcan A. *Paper-Based Multiplexed Vertical Flow Assay for Point-of-Care Testing.* **Lab on a Chip, 2019.**